A Survey on Self-Supervised Learning for Brain MRI Analysis: Robust Representation Learning Under Domain Shift

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***Abstract*—Brain Magnetic Resonance Imaging (MRI) is a cornerstone of neuro-oncological diagnosis and management, yet the development of robust Artificial Intelligence (AI) models remains challenged by limited expert annotations and pronounced domain shifts induced by multi-center heterogeneity. Variations in scanner hardware, acquisition protocols, and patient populations introduce substantial distributional discrepancies that severely impair the generalization of conventional supervised learning methods. In this context, Self-Supervised Learning (SSL) has recently attracted considerable attention as a paradigm for learning transferable representations from large-scale unlabeled brain MRI datasets. This survey provides a structured and critical review of recent advances in SSL for brain MRI analysis, with a particular focus on robustness to domain shift and inter-center generalization. We propose a concise taxonomy of SSL paradigms tailored to brain MRI, encompassing contrastive, reconstruction-based, generative, multimodal, and volumetric approaches, and analyze how these methods promote domain generalization through invariant yet task-relevant representations. We further discuss emerging trends such as 3D and multimodal SSL frameworks, hybrid pretraining objectives, and large-scale multi-domain pretraining strategies. Finally, we highlight key open challenges related to cross-site evaluation, computational scalability, interpretability, and the trade-off between invariance and diagnostically relevant variability, and outline future research directions, including causal and federated SSL.**

***Index Terms*—Self-supervised learning, brain MRI, domain generalization, domain shift, representation learning, medical image analysis**

I. Introduction

Neurological disorders constitute a major global health challenge, representing the leading cause of disability and the second leading cause of mortality worldwide, with more than 11 million deaths annually [1], [2]. Within this spectrum, cancers of the Central Nervous System (CNS) are among the most lethal conditions, with approximately 321,000 new cases and 248,000 deaths reported globally in 2022 [3]. This high mortality rate highlights the critical need for accurate and robust diagnostic and prognostic tools.

Machine Learning (ML) and Deep Learning (DL) have enabled major advances in many sectors, from manufacturing [4] to healthcare , but their impact is perhaps most profound in medicine. In particular, AI-assisted medical imaging is transforming patient care by enabling earlier detection and more personalized management [5]. This is especially evident in neuro-oncology, where brain MRI provides rich and detailed data, ideally suited for advanced analysis [6]. DL methods have now set new benchmarks for automated tumor segmentation, grading, and classification of brain diseases. However, their clinical translation remains limited by two fundamental challenges. First, supervised learning relies heavily on large-scale expert annotations, which are costly and scarce in neuroimaging. Second, substantial variability across scanners and acquisition protocols induces severe domain shifts, significantly degrading model generalization.

Self-Supervised Learning (SSL) has recently emerged as a promising paradigm to address these limitations. By exploiting unlabeled data through pretext tasks, SSL may enable the learning of transferable and data-efficient representations, reducing annotation dependency and improving robustness to cross-site variability [7]. In medical imaging, and particularly in brain MRI, SSL has demonstrated strong potential for enhancing domain generalization and downstream task performance.

In this survey, we provide a comprehensive and structured

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review of SSL methods for brain MRI analysis, with a particular focus on their role in mitigating domain shift.

The main contributions of this survey are threefold:

– **Compact taxonomy:** We propose a principled taxonomy of SSL paradigms tailored to brain MRI.
– **Application analysis:** We analyze SSL methods across major brain MRI tasks from a representation learning perspective.
– **Future directions:** We discuss current limitations and outline emerging research directions toward robust and generalizable neuroimaging models.

The remainder of this paper is organized as follows: Section II reviews brain MRI acquisition principles, sources of domain shift with a brief introduction to SSL. Section III presents the proposed SSL taxonomy. Section IV discusses emerging trends and open challenges, and Section V concludes the paper.

II. Technical Background

This section establishes the technical basis of this work, addressing standard brain MRI modalities and the main tumor characteristics. It then discusses the challenge of domain shift in multi-site studies and presents SSL as a method for learning robust representations from unlabeled medical data.

A. Brain MRI Modalities

Structural brain MRI relies on several complementary acquisition sequences commonly used in brain tumor analysis, each emphasizing distinct tissue properties [8]. The main modalities generally used are as follows:

- **T1-weighted (T1W):** Provides detailed anatomical structure, with clear white–gray matter contrast; commonly used for morphometry and structural segmentation.
- **T1-weighted contrast-enhanced (T1c):** Acquired after a contrast agent injection, highlights blood–brain barrier disruption and enhances tumor regions.
- **T2-weighted (T2W):** Sensitive to water content, facilitating the detection of edema and infiltrative abnormalities.
- **FLAIR:** Suppresses cerebrospinal fluid signal while preserving T2 contrast, improving lesion visibility near ventricles and white matter.

The visual differences between these sequences and their respective intensities are illustrated in Figure 1, which showcases the contrast variations across the four modalities.

B. Tumor Characteristics in MRI

Accurate characterization of brain tumors from MRI depends on understanding several critical factors related to tumor structure and imaging techniques. These factors include:

- **Heterogeneity of brain tumors:** Brain tumors exhibit substantial heterogeneity both within a single tumor and across patients, resulting in highly variable MRI appearances. Differences in tumor grade, subtype, and spatial structure contribute to complex imaging patterns that challenge automated analysis. This variability makes it difficult for supervised models trained on limited datasets to generalize across cases. Consequently, robust representation learning strategies are required to capture both local and global structural patterns in brain MRI [10].

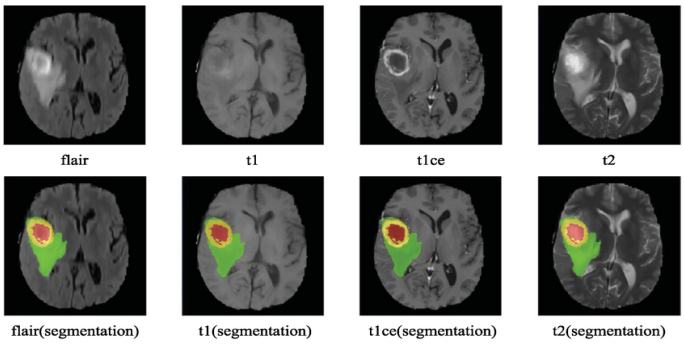


Fig. 1: Multimodal Brain MRI for Tumor Characterization [9]

- **Tumor subregions: Core, edema, and necrosis:** The visual characteristics and contrast variations of these four modalities are illustrated in Figure 2, which provides a comparative view of the same brain region across different sequences.

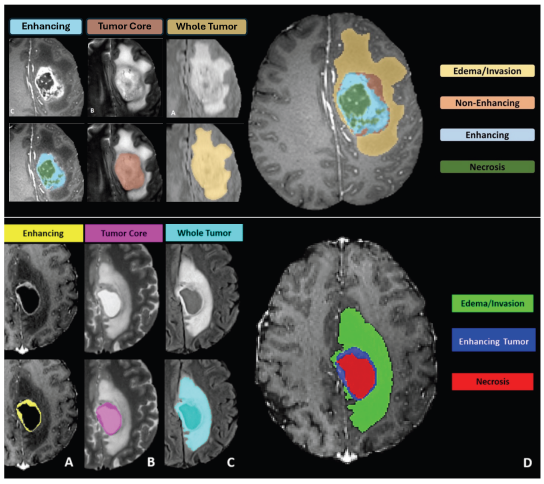


Fig. 2: Brain Tumor Segmentation and Sub-regions [9]

- **Multimodal relevance in MRI-based tumor analysis:** Multimodal MRI plays an important role in brain tumor analysis since different sequences provide complementary information about tissue characteristics. The T1-W MRI provides anatomical context, while T2-W and FLAIR MRI highlight abnormal tissue regions.
  From a ML perspective, combining multiple modalities can improve representation learning by integrating complementary features. However, multimodal datasets introduce challenges such as missing modalities and acquisition variability. SSL has recently been explored to exploit cross-modal relationships and learn robust representations that remain stable across heterogeneous MRI datasets [11].

C. Domain Shift in Brain MRI

The term domain shift refers to systematic discrepancies between data distributions originating from different imaging

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domains that are unrelated to the underlying clinical task of interest. In brain MRI, domain shift arises from a combination of scanner variability, acquisition protocol differences, and multi-site heterogeneity, and manifests as changes in intensity statistics, contrast patterns, noise characteristics, and spatial features. Variations in scanner vendors, magnetic field strengths, and radiofrequency coils introduce low-level appearance differences, while protocol-dependent parameters such as repetition time and slice thickness further amplify distributional shifts across datasets [12].

From an ML perspective, the domain shift constitutes an obstacle to model generalization. DL models trained on data from a single domain tend to capture domain-specific signatures rather than invariant anatomical or pathological representations, leading to significant performance degradation when deployed on previously unseen domains, such as MRI data acquired at different institutions or using different scanners. Importantly, several studies have shown that these domain-specific signatures remain detectable even after standard preprocessing steps, including skull stripping, spatial normalization, and intensity normalization, highlighting the intrinsic nature of domain shift in neuroimaging [13].

Overall, domain shift represents an intrinsic characteristic of multi-center neuroimaging datasets. Variations in scanners, acquisition protocols, and acquisition environments introduce systematic discrepancies in MRI appearance that are unrelated to the underlying clinical task. Consequently, models trained on data from a limited number of sites may capture domain-specific patterns rather than stable anatomical structures, leading to reduced reliability when applied to data acquired under different imaging conditions [14].

These variations in acquisition parameters and their impact on image appearance are visually demonstrated in Figure 3, where shifts in intensity and contrast are evident across different scanner settings.

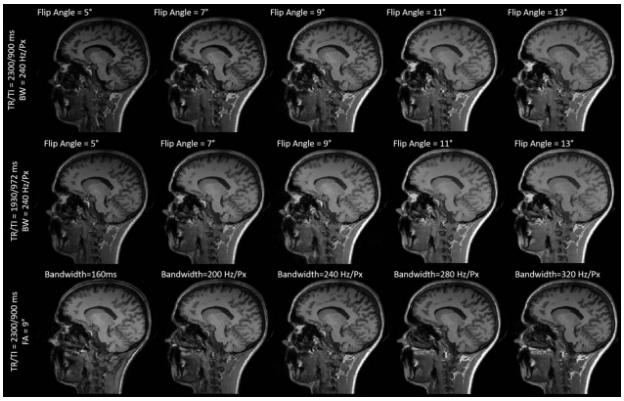


Fig. 3: Domain Shift Analysis [15]

D. Self Supervised Learning

SSL has recently attracted significant attention as a promising paradigm for representation learning in medical imaging, particularly in brain MRI, where expert annotations are scarce, costly, and subject to inter-observer variability. Unlike conventional supervised learning, SSL leverages large amounts of unlabeled data by defining pretext tasks that automatically generate learning signals. This strategy may enable the learning of robust and transferable representations that are well suited for clinical tasks such as classification, segmentation, and anomaly detection. Recent studies have demonstrated that self-supervised pretraining significantly improves performance, especially under low-supervision regimes and in the presence of domain shift across imaging centers and acquisition protocols [16], [17].

Formally, each MRI image x_i is transformed into a latent representation z_i via an encoder f_θ:

$$z_i = f_\theta(x_i), \quad x_i \in \mathbb{R}^{H\times W\times D}, \quad z_i \in \mathbb{R}^d \tag{1}$$

This latent representation serves as the foundation for all SSL tasks.

The fundamental principle of SSL relies on the design of pretext tasks, such as image reconstruction, transformation prediction, spatial context modeling, or masked region modeling. These mechanisms encourage models to capture anatomical coherence, structural relationships, and biological variability inherent to MRI data. In generative approaches, the model learns to reconstruct the original image x_i from a corrupted version $\tilde{x}_i$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - g_\phi(f_\theta(\tilde{x}_i))\|_2^2 \tag{2}$$

where g_ϕ denotes the decoder and $\mathcal{L}_{\text{recon}}$ is the reconstruction loss.

Data augmentation strategies also play a central role by generating multiple views of the same volume, thereby promoting invariance to noise, intensity variations, and inter-scanner differences. This ability to learn invariant representations is particularly critical in real-world clinical environments characterized by strong heterogeneity [16], [18], [19].

SSL methods can be broadly grouped into generative, contrastive, predictive, and non-contrastive approaches. Generative methods focus on data reconstruction and distribution modeling, while contrastive learning aims to maximize similarity between different views of the same sample.

The conceptual framework of this methodology, depicting the transition from SSL-based feature initialization using unlabeled MRI data to the subsequent transfer for clinical applications, is illustrated in Figure 4.

For contrastive learning, two augmented views of the same sample, x_i^a and x_i^b, are created, and the model is trained to bring their latent representations z_i^a and z_i^b closer while separating negatives:

$$\mathcal{L}_{\text{contrastive}} = -\sum_{i=1}^{N} \log \frac{\exp(\text{sim}(z_i^a, z_i^b)/\tau)}{\sum_{j=1}^{2N} \mathbf{1}_{[j\neq i]} \exp(\text{sim}(z_i^a, z_j)/\tau)} \tag{3}$$

where $\text{sim}(u, v) = \frac{u^\top v}{\|u\|\|v\|}$ is the cosine similarity and τ is a temperature hyperparameter [20].

More recently, non-contrastive and masking-based approaches have gained popularity due to their improved stability

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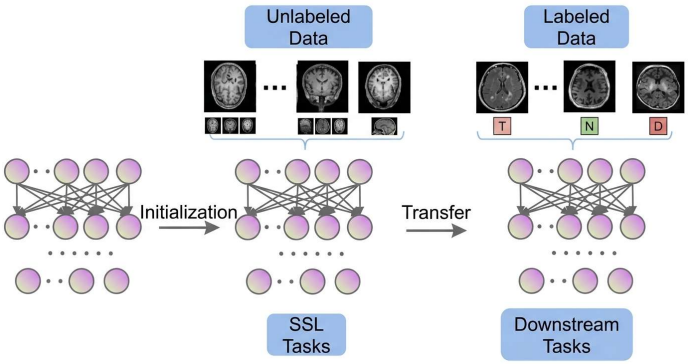


Fig. 4: SSL pipeline: pre-training on unlabeled data followed by fine-tuning or evaluation on labeled downstream tasks.

and scalability. These methods have demonstrated promising performance in brain MRI analysis, particularly for tumor segmentation, lesion detection, and the prediction of neurodegenerative diseases [21], [22].

In masked region modeling, certain regions $\mathcal{M}_i$ of the image are masked, and the model learns to predict the missing values:

$$\mathcal{L}_{\text{mask}} = \frac{1}{|\mathcal{M}_i|} \sum_{p \in \mathcal{M}_i} \|x_i[p] - \hat{x}_i[p]\|_2^2 \tag{4}$$

where $\hat{x}_i[p] = g_\phi(f_\theta(\tilde{x}_i))$ is the predicted value at position p.

Despite these advances, several challenges remain, including the three-dimensional nature of MRI data, inter-site generalization, robustness to domain shifts, and the clinical interpretability of learned representations. Consequently, SSL has recently attracted increasing attention in medical imaging as a strategy for learning transferable representations from large-scale unlabeled datasets.

In practice, these different losses can be combined into a multi-objective SSL loss:

$$\mathcal{L}_{\text{SSL}} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{contrastive}}\mathcal{L}_{\text{contrastive}} + \lambda_{\text{mask}}\mathcal{L}_{\text{mask}} \tag{5}$$

The coefficients λ allow balancing the contribution of each task, enabling the model to learn robust representations.

III. Taxonomy of Self-Supervised Learning in Brain MRI

In this survey, we propose a principled taxonomy of SSL for brain MRI, organized by learning principles and internal supervisory signals rather than architectures, clarifying how SSL enables robust representations under domain shift.

Our SSL taxonomy for neuroimaging, depicted in Figure 5, organizes methods by their supervisory signals. This framework illustrates a progression from local, voxel-level reconstruction tasks, such as masked autoencoding, to feature-level global methods, including SimCLR and multimodal alignment. This evolution reflects a trend toward more robust representations, incorporating physical principles, which improve invariance to scanner artifacts and protocol variations, leading to better cross-site generalization. In the following sections, we detail each paradigm in this taxonomy.

A. Voxel-Level Signal: Generative Learning Principle

Voxel-level SSL methods rely on the raw intensity values of MRI voxels as the primary source of supervision. The central learning principle is generative: the model is trained to reconstruct missing, corrupted, or masked regions of the input volume, thereby approximating the underlying data distribution [23], [24], [25], [18].

The supervisory signal arises from local spatial correlations between neighboring voxels, encouraging the network to capture fine-grained anatomical and textural patterns. Representative approaches include contextual inpainting, denoising autoencoders, and more recently, three-dimensional masked autoencoders tailored to volumetric MRI data.

This learning principle is particularly well suited to brain MRI, where subtle intensity variations encode critical anatomical information. By forcing the model to reconstruct cortical folds, ventricular boundaries, or lesion textures, voxel-level generative SSL promotes a detailed understanding of local brain structure. However, because the supervision is largely local, these methods may be sensitive to scanner-specific intensity distributions, which motivates their combination with more global learning principles [26].

Representative approaches in this category include denoising autoencoders, Variational AutoEncoders (VAEs), and masked autoencoding frameworks such as Masked AutoEncoders (MAE) [27]. In these approaches, the model learns to reconstruct the original input from a corrupted or partially observed version of the data.

Recent studies have extended these architectures to 3D medical imaging by designing 3D masked autoencoders capable of operating directly on MRI volumes [28]. These models randomly mask portions of input data and train network to reconstruct the missing regions. By solving this reconstruction task, the encoder learns representations that capture spatial continuity and structural organization within the brain.

However, purely reconstruction-based objectives may encourage the model to focus on low-level intensity statistics rather than high-level semantic features [29]. As a result, representations learned through voxel-level reconstruction may remain sensitive to scanner-specific intensity distributions and acquisition artifacts. For this reason, several recent studies have proposed hybrid approaches that combine generative objectives with contrastive or predictive learning signals to encourage more semantically meaningful representations [30]. In practice, voxel-level generative SSL is frequently used as a pretraining strategy for downstream tasks such as brain tumor segmentation, MRI reconstruction, and representation learning for volumetric neuroimaging analysis.

B. Feature-Level Signal: Contrastive Learning Principle

The key signal exploited in this paradigm is semantic invariance: a brain MRI scan should map to a similar representation despite variations induced by data augmentation, noise, or acquisition conditions. Popular frameworks such as contrastive predictive coding, momentum-based contrastive

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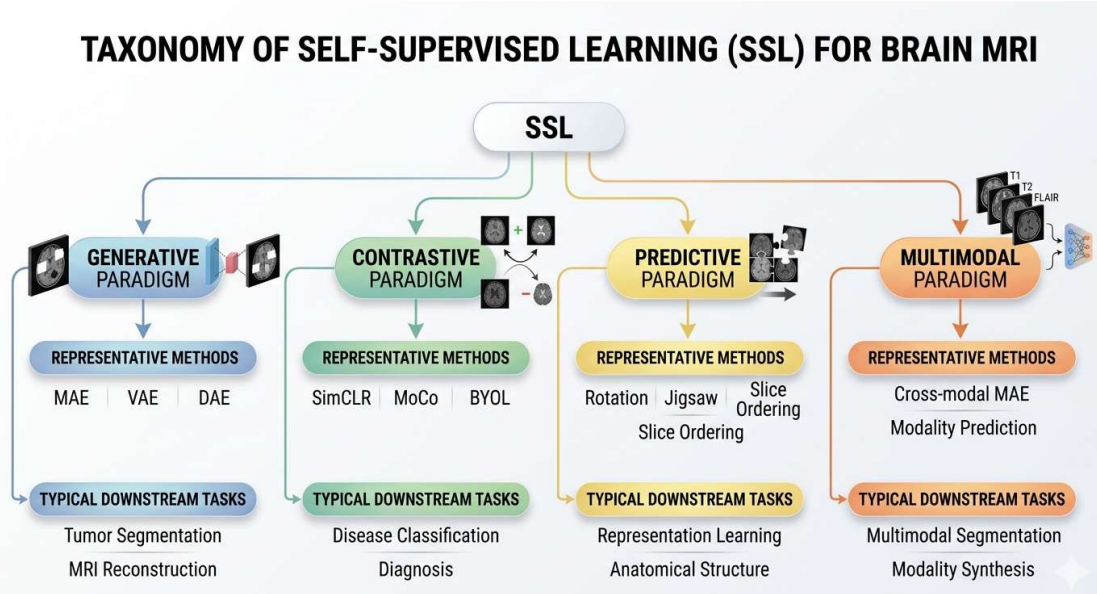


Fig. 5: Taxonomy of SSL Paradigms in Brain MRI. The progression from left to right illustrates the shift from local reconstruction tasks to global, physics-informed representations that enhance cross-site generalization.

learning, and bootstrap-based methods instantiate this principle using similarity-based objectives.

For brain MRI, contrastive learning is particularly effective in mitigating domain shift, as it encourages representations that are invariant to scanner-specific artifacts. Moreover, unlike natural image analysis, brain MRI offers unique opportunities for defining meaningful positive pairs, such as longitudinal scans of the same patient acquired at different time points. This makes feature-level contrastive learning a powerful tool for learning global, subject-level representations that generalize across centers [21], [17], [22].

Contrastive frameworks include SimCLR [31], Momentum Contrast (MoCo), and Bootstrap Your Own Latent (BYOL). These approaches learn representations by maximizing the agreement between different augmented views of the same input sample while simultaneously pushing apart representations of different samples [20].

Modern contrastive frameworks are frequently implemented using convolutional neural networks or vision transformer architectures that encode MRI volumes into compact latent representations. Projection heads and memory queues are often incorporated to stabilize training and improve the quality of the learned embeddings when large-scale datasets are available [32], [33].

Contrastive learning has been particularly effective for learning global semantic representations of brain MRI volumes. However, its performance strongly depends on the design of data augmentations and the availability of sufficiently large training datasets. In medical imaging scenarios where data are limited or heterogeneous, selecting appropriate augmentation strategies remains a critical challenge. Feature-level contrastive learning has been widely explored for downstream classification tasks, including disease diagnosis, tumor grading, and clinical outcome prediction from brain MRI data.

C. Geometric and Structural Signal: Predictive Learning Principle

Predictive SSL methods exploit the intrinsic geometric and structural properties of brain MRI data without relying on explicit reconstruction. Instead, the model is trained to predict spatial or contextual attributes derived from the image itself, such as orientation, relative patch location, or slice ordering [16], [34].

The supervisory signal in this category is based on spatial metadata and anatomical regularities. Because the human brain exhibits a highly organized and symmetric structure, predicting the relative position of image regions implicitly forces the model to learn global anatomical context and spatial coherence. This learning principle is particularly relevant for volumetric MRI, where understanding three-dimensional relationships between slices is essential. By leveraging geometric consistency, predictive SSL methods encourage representations that encode anatomical layout rather than appearance alone, which contributes to improved robustness when transferring across datasets with different intensity profiles or acquisition protocols [35].

Predictive SSL methods rely on auxiliary tasks that require the model to infer structural properties of the input data. Typical predictive tasks include rotation prediction, relative patch location prediction, jigsaw puzzle reconstruction, and slice ordering prediction in volumetric MRI data.

Because these tasks rely on spatial reasoning rather than direct intensity reconstruction, predictive SSL methods can learn representations that emphasize anatomical structure rather than scanner-specific intensity patterns. This property may

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contribute to improved robustness when models are applied across datasets acquired with different scanners or imaging protocols. Nevertheless, the effectiveness of predictive tasks depends on how well the auxiliary objective aligns with the downstream clinical application.

Predictive SSL approaches have been investigated to improve anatomical segmentation, structural landmark detection, and spatial understanding in volumetric brain MRI analysis.

D. Multi-Modal and Physics-Based Signal: Cross-Sequence Learning Principle

Multi-modal SSL exploits redundancy and complementarity between different MRI sequences, such as T1W, T2W, and FLAIR images. Unlike voxel or feature-level methods, the supervisory signal here is rooted in the physics of MRI acquisition, as different sequences represent distinct contrast mechanisms of the same underlying anatomy [36]. The learning principle consists in predicting one modality from another or enforcing consistency across modalities within a shared latent space. By doing so, the model is encouraged to disentangle anatomical information from contrast-specific characteristics.

This category is particularly important for addressing domain shift in multi-center studies, where sequence availability and acquisition parameters vary widely. Cross-sequence SSL promotes contrast-invariant representations that remain stable across scanners and protocols, making it a key paradigm for clinical generalization and deployment [37], [38], [39].

Recent work has explored cross-modal self-supervision, where models learn shared representations by predicting one MRI modality from another or by aligning modality-specific embeddings within a common latent space [40], [41].

Transformer-based architectures have recently been adopted to model relationships between modalities in large-scale neuroimaging datasets. By leveraging attention mechanisms, these models can capture long-range dependencies and interactions between different imaging modalities.

Multimodal SSL approaches may require carefully aligned multimodal datasets and can be sensitive to missing modalities, which remain common challenges in clinical neuroimaging studies [40]. Cross-modal SSL methods are particularly relevant for multimodal tumor segmentation, modality synthesis, and integrative analysis of heterogeneous neuroimaging datasets.

Voxel-level SSL methods rely on the raw intensity values of MRI voxels as the primary source of supervision. The central learning principle is generative: the model is trained to reconstruct missing, corrupted, or masked regions of the input volume, approximating the underlying data distribution.

Table I summarizes the main categories of SSL methods used in brain MRI analysis, together with representative architectures and their typical downstream clinical applications.

IV. Emerging Trends, Challenges, and Future Directions

Recent advances in SSL have profoundly transformed brain MRI analysis. Beyond achieving improved performance on individual tasks, contemporary research increasingly prioritizes robustness, scalability, and generalization across heterogeneous imaging domains. Recent research has focused on improving the scalability, robustness, and anatomical consistency of SSL-based representation learning frameworks for brain MRI analysis [17]. This section surveys emerging trends in SSL for brain MRI, examines its role in promoting domain generalization, discusses outstanding challenges, and highlights promising directions for future research.

A. Emerging Trends in SSL for Brain MRI

Key emerging trends in SSL for brain MRI reflect a shift toward more aware, multimodal, and domain-robust representation learning paradigms. These trends include:

- **Transition from 2D to 3D SSL frameworks:** Given the volumetric nature of brain MRI, 3D SSL captures long-range spatial dependencies better than slice-based 2D methods, producing more anatomically consistent representations despite higher computational costs [17].
- **Multimodal and cross-modal representation learning:** These methods jointly exploit complementary MRI sequences within a shared latent space to enhance representation richness and robustness, while mitigating performance degradation from missing modalities commonly encountered in clinical practice [19].
- **Domain generalization–oriented SSL:** Robustness to scanner, protocol, and site-specific variations is treated as a primary design objective. By leveraging large-scale, diverse, unlabeled multi-center data during pretraining, these approaches explicitly reduce sensitivity to domain-specific biases and improve generalization across heterogeneous imaging environments [17].

B. Domain Generalization through SSL

Key aspects of domain generalization through SSL in brain MRI include:

- **Distinction between invariance and robustness:** Invariance suppresses task-irrelevant variations, while robustness preserves diagnostically meaningful ones. Excessive invariance may eliminate subtle cues critical for diagnosis, making careful SSL design crucial [19].
- **Balancing task-relevant variability and domain suppression:** A central challenge in SSL design for neuroimaging is achieving an optimal trade-off between reducing domain-specific biases and retaining task-relevant variability. Effective SSL frameworks must preserve clinically informative features while minimizing sensitivity to irrelevant scanner or protocol differences [14].
- **Limitations of current evaluation practices:** Reliance on intra-dataset splits or limited cross-validation underestimates real-world deployment challenges, yielding overly optimistic performance estimates that fail to capture true generalization to unseen institutions or scanners. [44].
- **Need for rigorous multi-center evaluation protocols:** Accurate assessment of domain generalization requires multi-center, cross-scanner, and cross-protocol evaluation

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TABLE I: Overview of self-supervised learning paradigms for brain MRI and their typical downstream applications

SSL Paradigm	Representative Methods	Architecture	Typical Downstream Tasks	Notes	Research Papers
Generative	MAE, VAE, DAE	Autoencoders, Vision Transformers (ViT)	Tumor segmentation, MRI reconstruction	Focuses on voxel-level detail recovery	[18], [23], [24], [25], [28], [30]
Contrastive	SimCLR, MoCo, BYOL	CNN, Vision Transformers	Disease classification, diagnosis	Learns high-level invariant features	[17], [21], [22], [42]
Predictive	Rotation, jigsaw, slice ordering	CNN	Representation learning, anatomical understanding	Leverages spatial/structural cues	[16], [34], [43]
Multimodal	Cross-modal MAE, modality prediction	Transformer-based architectures	Multimodal tumor segmentation, synthesis	Exploits cross-sequence correlations	[37], [38], [39], [40], [41]

settings. Only through such rigorous protocols can the real-world robustness of SSL-pretrained models be reliably quantified [17].

C. Open Challenges

Despite substantial progress, several challenges hinder the widespread adoption and clinical translation of SSL-based brain MRI models. Key challenges include:

- **Lack of standardized benchmarks:** The diversity of pretext tasks, network architectures, and evaluation strategies complicates fair comparison across methods and limits its reproducibility. Standardized benchmarks specifically tailored for SSL are still scarce, making it difficult to compare different approaches in a consistent manner [19].
- **Limited multi-center validation:** Most SSL models are assessed on only a few datasets, raising concerns about their robustness in diverse clinical settings. Large-scale multi-center studies are essential to ensure reliable performance across different scanners and protocols [17].
- **High computational cost:** Advanced 3D and multimodal SSL frameworks require substantial memory resources and long training times, which limits scalability and practical deployment in real-world clinical settings [17].
- **Handling missing or incomplete modalities:** Real-world multimodal datasets rarely adhere to ideal assumptions, with missing or incomplete modalities being common. SSL frameworks need to robustly handle such scenarios to be useful in clinical workflows [41].

D. Future Research Directions

Future research in SSL for brain MRI is expected to move toward more principled and scalable paradigms. One promising direction is the development of domain-invariant and causal SSL approaches that explicitly disentangle pathology-related factors from acquisition-specific variations. By incorporating causal reasoning into representation learning [19].

Another emerging avenue is federated SSL, which may enable collaborative representation learning across institutions without sharing raw imaging data, thereby addressing privacy and regulatory constraints. In parallel, the rise of large-scale foundation models pretrained on diverse neuroimaging datasets opens new opportunities for universal representations transferable across tasks, modalities, and institutions [17].

Finally, integrating SSL with explainability mechanisms is essential for fostering clinical trust and adoption. Combining SSL with interpretable modeling strategies may bridge the gap between high-performance representation learning and transparent, clinically actionable decision-support systems [41].

V. Conclusion

This survey has provided a comprehensive overview of recent advances in SSL for brain MRI analysis, with emphasis on robust representation learning under domain shift. By reviewing contrastive, reconstruction-based, generative, multimodal, and volumetric SSL paradigms, we highlighted how SSL addresses both annotation scarcity and domain variability in neuro-oncology. Across tasks including tumor segmentation, classification, reconstruction, and modality synthesis, SSL-based approaches demonstrate improved data efficiency and generalization compared to supervised models. Pretraining on large-scale, heterogeneous multi-center MRI datasets enables models to learn representations less sensitive to scanner- and protocol-specific variations, underscoring the importance of strategies that account for neuroimaging data heterogeneity.

However, clinical adoption remains limited by the lack of standardized benchmarks, insufficient cross-site validation, high computational costs, and limited interpretability. Balancing invariant feature learning with clinically meaningful variability also remains an open challenge.

Future progress will depend on domain-invariant and causally grounded SSL methods, federated learning, and large-scale foundation models, alongside integration with explainability and clinically validated pipelines to bridge the gap between innovation and real-world impact.

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